

Production Test Certificate

General Information

WIP Job No:	13437903	Customer Order No:	UNASSIGNED
Customer Name:	UNASSIGNED	Test Cell No:	5

Product Information

Model:	C450 D6E	Genset Serial No:	J15K879647
Engine Serial No:	79865137	Alternator Serial No:	A15I399191
Fuel Type:	DIESEL	Controller Type:	PC2.2
Governor Type:	Electronic	Radiator Serial No:	N/A
Cooling System Type:	Mounted Radiator	Controller Serial No:	0065085D14

Test Conditions

Altitude (Meters ASL):	32	Barometric Pressure(mB):	1017
Ambient Temp(C):	16	Humidity(%):	29
Fuel Oil Spec:	BS 2869 Class A2	Lub Oil Spec:	Premium Blue 15W40
Fuel Density(kg/m):	820 - 875	Fuel Calorific Value:	43500 kJ/kg
Air Filter Inlet Temp(C):	16	Coolant Mix:	50/50 - Ethylene Glycol/Water

Product Options Checked

☒ Emergency Stop	Panel Heater	☒ Low Coolant S/Down	Low Fuel S/Down	High Fuel Warning
☒ Remote Start	Alternator Heater	Low Coolant Warning	Low Fuel Warning	Parallel Bus PT Module
Coolant Heater	Run Relay	☒ LET Warning	High Fuel S/Down	Earth Fault S/Down
CCB Trip Warning				

Technical Data - Declared

Service Rating:	STANDBY	Power Factor:	0.8
Power Output(kW):	450	Rated Output(kVA):	562
Rated Voltage(V):	380	Rated Current(A):	854
Frequency(Hz):	60	Speed(RPM):	1800
Number Of Phases:	3	Performance Class:	G2
Meter Serial No:	16071		

Technical Data - Measured

Results(No Load)

Voltage:	1-2:	383	Volts	2-3:	383	Volts	1-3:	383	Volts
	1-N:	221	Volts	2-N:	221	Volts	3-N:	221	Volts
No Load Freq:		60	Hz	No Load Coolant Temp(C): 75					
No Load Lube Press:		3.4	Bar						

Results(Full Load)

Voltage:	1-2:	380	Volts	2-3:	380	Volts	1-3:	380	Volts
	1-N:	220	Volts	2-N:	220	Volts	3-N:	220	Volts
Current:	1:	684	Amps	2:	684	Amps	3:	684	Amps
Full Load Frequency:		60	Hz	Full Load Coolant Temp(C):	79	Power Factor:	1.0		
Full Load Lube Press:		3	Bar	Prime set tested to 110% of Full Load(if r'eqd):	N/A				

PTI/Static Test Completed By:	Pete Ayles	Date:	09-OCT-15	
Dynamic Test Completed By:	Norman Phillips	Date:	09-OCT-15	Run Hours: 0.65

Technical File: Other Standards Applied

Machinery Directive, Directive 2006/42/EC: -	
SI 2008 No. 1597	The Supply of Machinery (Safety) Regulations 2008.
SI 2002 No. 1144	The Personal Protective Equipment Regulations 2002
BS EN 626-1:1994+A1:2008	Safety of machinery – Reduction of risks to health from hazardous substances emitted by machinery – Part 1: Principals and specifications for machinery manufacturers.
BS EN 626-2:1996+A1:2008	Safety of machinery – Reduction of risks to health from hazardous substances emitted by machinery – Part 2: Methodology leading to verification process.
BS EN ISO 3745:2012	Determination of sound power levels and sound energy levels of noise sources using sound pressure - Precision methods for anechoic rooms and hemi-anechoic rooms
BS EN ISO 3746:2010	Acoustics. Determination of sound power levels of noise sources using sound pressure. Survey method using an enveloping measurement surface over a reflecting plane
BS ISO 8528-1:2005	Reciprocating internal combustion engine driven alternating current generating sets. Application, ratings and performance.
BS ISO 8528-2:2005	Reciprocating internal combustion engine driven alternating current generating sets. Engines.
BS ISO 8528-3:2005	Reciprocating internal combustion engine driven alternating current generating sets. Alternating current generators for generating sets.
BS ISO 8528-4:2005	Reciprocating internal combustion engine driven alternating current generating sets. Controlgear and switchgear.
BS ISO 8528-5:2005	Reciprocating internal combustion engine driven alternating current generating sets. Generating sets.
BS ISO 8528-6:2005	Reciprocating internal combustion engine driven alternating current generating sets. Test methods.
BS 7698-7:1996, ISO 8528-7:1994	Reciprocating internal combustion engine driven alternating current generating sets. Technical declarations for specification and design
BS 7698-8:1996, ISO 8528-8:1995	Reciprocating internal combustion engine driven alternating current generating sets. Requirements and tests for low-power generating sets
BS 7698-9:1996, ISO 8528-9:1995	Reciprocating internal combustion engine driven alternating current generating sets. Measurement and evaluation of mechanical vibrations
BS 7698-10:1999, ISO 8528-10:1998	Reciprocating internal combustion engine driven alternating current generating sets. Measurement of airborne noise by the enveloping surface method
BS 7698-12:1998, ISO 8528-12:1997	Reciprocating internal combustion engine driven alternating current generating sets. Emergency power supply to safety devices

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LV Directive 2006/95/EC.

Standard Number	Title
BS EN 60204-1:2006+A1:2009.	Safety of machinery — Electrical equipment of machines — Part 1: General requirements.
BS EN 60439-1:1999, IEC 60439-1:1999	Low voltage switchgear and controlgear assemblies — Part 1: Type-tested assemblies.
BS EN 60446:2007.	Basic and safety principles for man-machine interface, marking and identification — Identification of conductors by colours or alphanumerics.
BS EN 61439-1:2011	Low-voltage switchgear and controlgear assemblies. General rules
BS EN 61439-2:2011	Low-voltage switchgear and controlgear assemblies. Power switchgear and controlgear assemblies
BS EN 60529:1992+A2:2013	Degrees of protection provided by enclosures (IP code) (IEC 529:1989).
BS EN 60947-1:2007+A1:2011.	Low-voltage switchgear and controlgear. General rules.
BS EN 61310-1:2008.	Safety of machinery – Indication, marking and actuation – Part 1: Requirements for visual, auditory and tactile signals.)
BS EN 61310-2:2008.	Safety of machinery – Indication, marking and actuation – Part 2: Requirements for marking.

EMC Directive 2004/108/EC.

Standard Number	Title
BS EN 61000-6-1:2007.	Electromagnetic compatibility (EMC). Generic standards. Immunity for residential, commercial and light-industrial environments.
BS EN 61000-6-2:2005.	Electromagnetic compatibility (EMC). Generic standards. Immunity standard for industrial environments.
BS EN 61000-6-3:2007+A1:2011.	Electromagnetic compatibility (EMC). Generic standards. Emission standard for residential, commercial and light-industrial environments.
BS EN 61000-6-4:2007+A1:2011.	Electromagnetic compatibility (EMC). Generic standards. Emission standard for industrial environments.

Noise Directive 2000/14/EC as amended by 2005/88/EC.

Standard number	Title
BS EN ISO 3744:2010.	Acoustics. Determination of sound power levels and sound energy levels of noise sources using sound pressure. Engineering methods for an

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**Power
Generation**

	essentially free field over a reflecting plane.
BS ISO 8528-1:2005	Reciprocating internal combustion engine driven alternating current generating sets. Application, ratings and performance.
BS 7698-10:1999, ISO 8528-10:1998	Reciprocating internal combustion engine driven alternating current generating sets. Measurement of airborne noise by the enveloping surface method.